

COLLEGE KNOWLEDGE ASSISTANT

PROJECT REPORT

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Date: 06.09.2026

1. Problem Statement

Students frequently have questions regarding college policies, attendance, examinations, placements, hostel facilities and other academic information. Finding this information manually from different college documents can be time-consuming.

The objective of this project is to develop a **College Knowledge Assistant** that can answer college-related questions using information available in a knowledge base. The system uses **Retrieval Augmented Generation (RAG)** so that relevant information can be retrieved before generating an answer.

The assistant is designed to avoid providing unsupported college-specific information. If the required information is not available in the knowledge base, the system is instructed to respond that the information could not be found.

2. Architecture

The system consists of the following major components:

Student → AI Agent → Vector Store → OpenAI Chat Model → Response

Knowledge Ingestion

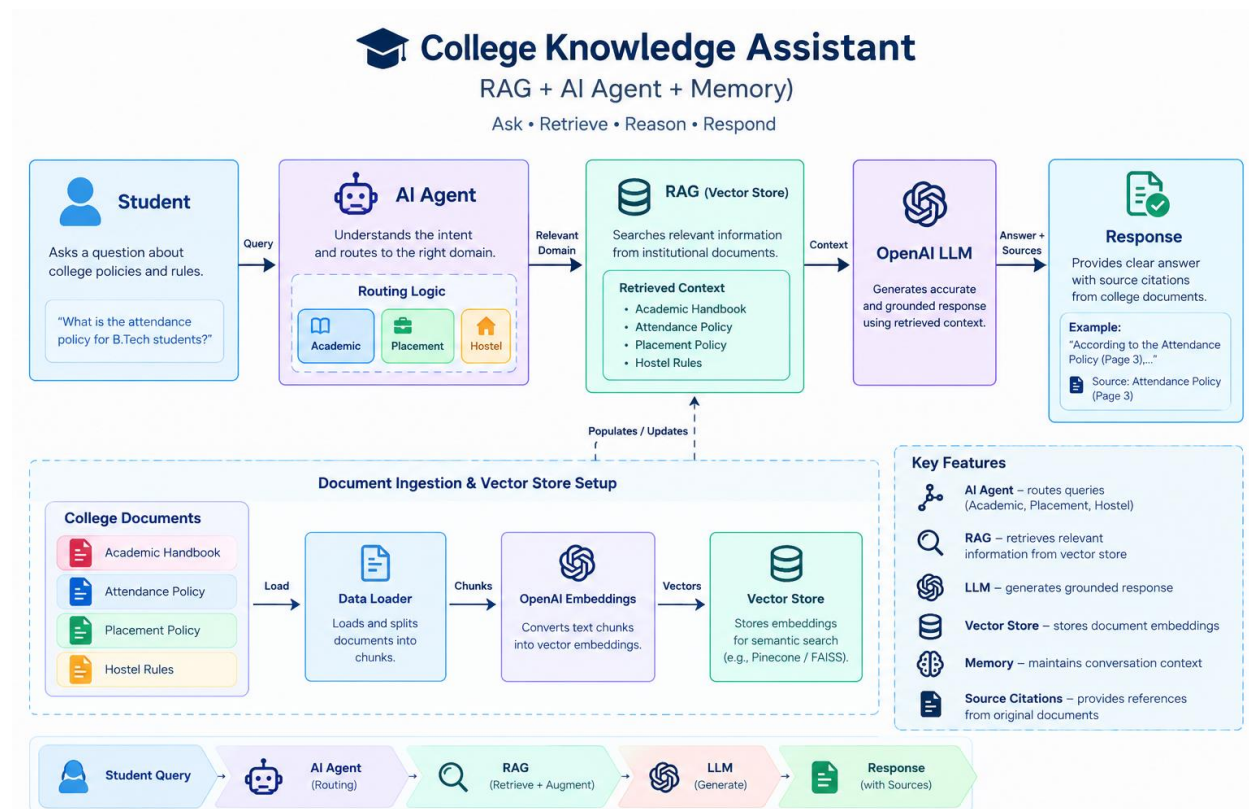
College information is first processed through a **Default Data Loader**. OpenAI Embeddings convert the information into vector representations, which are stored in the **Simple Vector Store**.

Question Answering

When a student asks a question, the **AI Agent** receives the question and uses **Simple Vector Store¹** as a retrieval tool. The vector store searches for relevant information and provides it to the AI Agent. The OpenAI Chat Model then generates the final response based on the retrieved information.

The workflow also contains **Simple Memory**, which maintains conversation context using a custom session key.

Architecture Diagram



3. Prompt Design

The AI Agent is configured with a system prompt specifically for the College Knowledge Assistant.

The prompt instructs the agent to:

- Use the college knowledge-base retrieval tool before answering college-related questions.
- Use the student's question as the search query.
- Avoid answering college-specific questions using unsupported information.
- Generate the answer using retrieved college documents.
- Provide a fallback response when the required information is not available.

The fallback instruction is:

“I couldn't find this information in the college knowledge base.”

This helps reduce unsupported or fabricated answers.

4. Implementation and Technologies

The project was implemented using **n8n** and its AI/LangChain components.

Technologies Used

- **n8n** – workflow automation
- **AI Agent** – handles questions and retrieval
- **OpenAI Chat Model** – generates responses
- **OpenAI Embeddings** – converts information into vector representations
- **Simple Vector Store** – stores and retrieves knowledge
- **Default Data Loader** – loads document content
- **Simple Memory** – maintains conversation context
- **Webhook** – provides an HTTP endpoint for the assistant

The configured workflow uses the `gpt-5-mini` OpenAI Chat Model.

5. Evaluation

The system was tested using college-related questions.

Test Case 1

Question:

“What percentage of attendance is required to be eligible for the final examination?”

Expected Answer:

75% attendance is required.

The attendance policy included in the workflow states that students must maintain at least 75% attendance to be eligible for the final examination.

Test Case 2**Question:**

“What happens if attendance is between 65% and 75%?”

Expected Behaviour:

The assistant should explain that students may be allowed to appear for examinations after approval from the academic department.

Test Case 3**Question:**

“What happens if attendance is below 65%?”

Expected Behaviour:

The assistant should explain the relevant attendance rule from the knowledge base.

Test Case 4**Question:**

“What is the admission process of the college?”

Expected Behaviour:

If the information is not available in the knowledge base, the assistant should use the configured fallback response rather than inventing an answer.

6. Challenges

During development, several challenges were encountered, including configuring the AI Agent, connecting the vector store with embeddings, setting up document loading and configuring conversation memory.

Another important challenge was ensuring that the AI Agent uses retrieved knowledge instead of relying on unsupported information. This was addressed by adding explicit instructions in the system prompt requiring the retrieval tool to be used for college-related questions.

7. Future Enhancements

The project can be further improved by:

1. Adding more official college documents to the knowledge base.
2. Adding dedicated academic, placement and hostel knowledge sources.
3. Improving retrieval accuracy using better chunking and retrieval strategies.
4. Adding clearer source references to responses.
5. Developing a user-friendly web or chat interface.
6. Adding authentication so that only authorized students can access the assistant.
7. Continuously evaluating the system using a larger set of real student questions.

8. Conclusion

The **College Knowledge Assistant** demonstrates how RAG, vector search, an AI Agent and a large language model can be combined to create a useful college information assistant.

The system retrieves relevant information from its knowledge base before generating answers and includes conversation memory for maintaining context. The project provides a foundation that can be expanded with additional college documents and a dedicated student-facing interface.

9. Demo Video

The complete working demonstration of the college knowledge assistant is available at the following link :

<https://drive.google.com/file/d/1hGEsAd21WFMp4foDy15vqTKrdt8k4gId/view?usp=sharing>